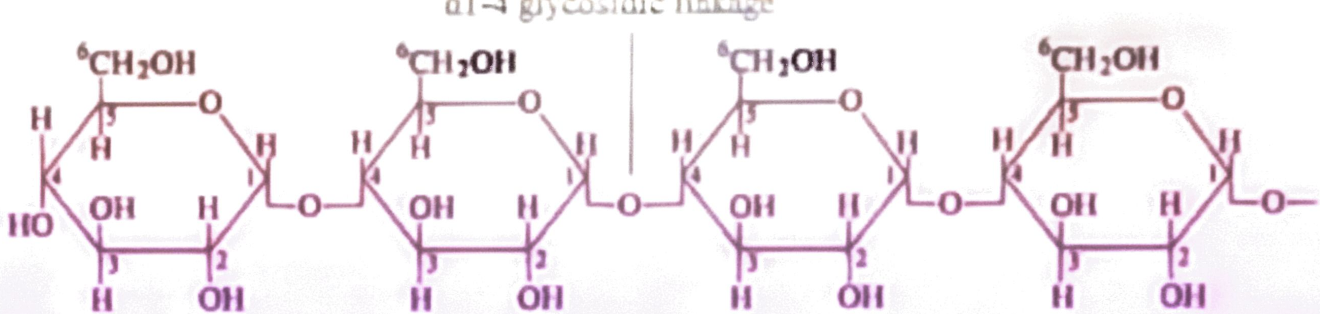
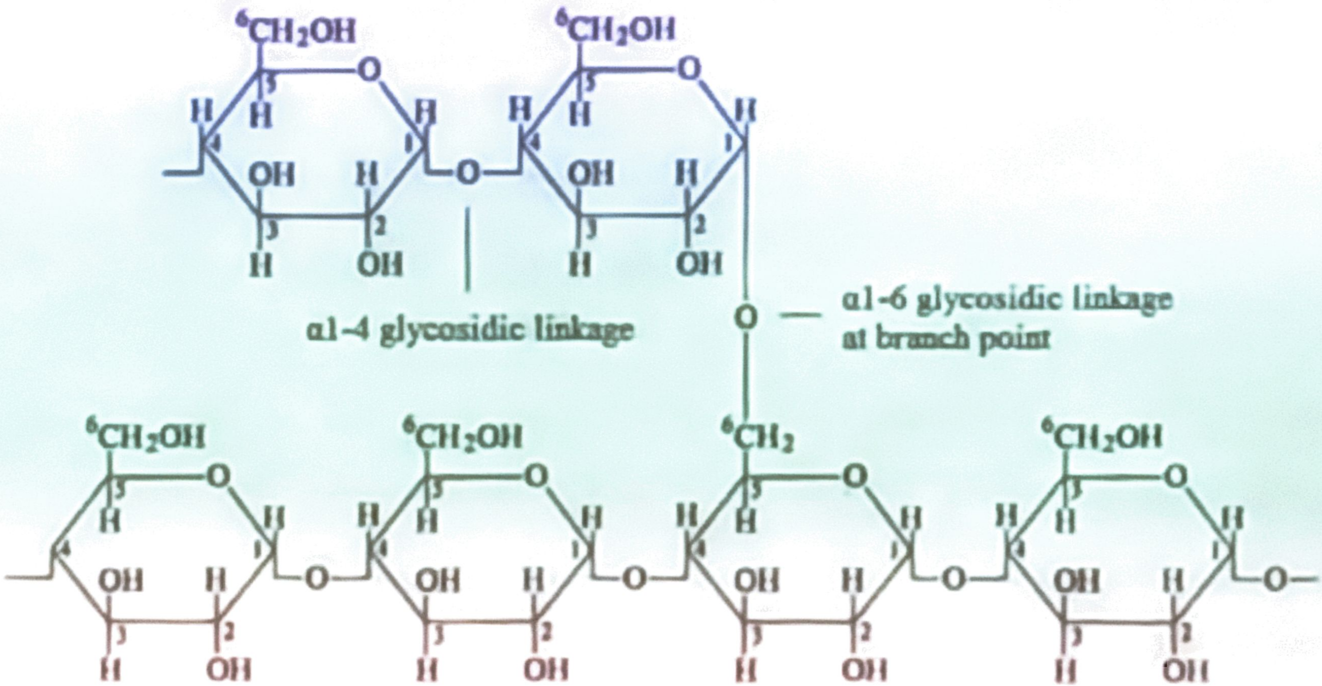
25.5 Homopolysaccharides: A. Starch

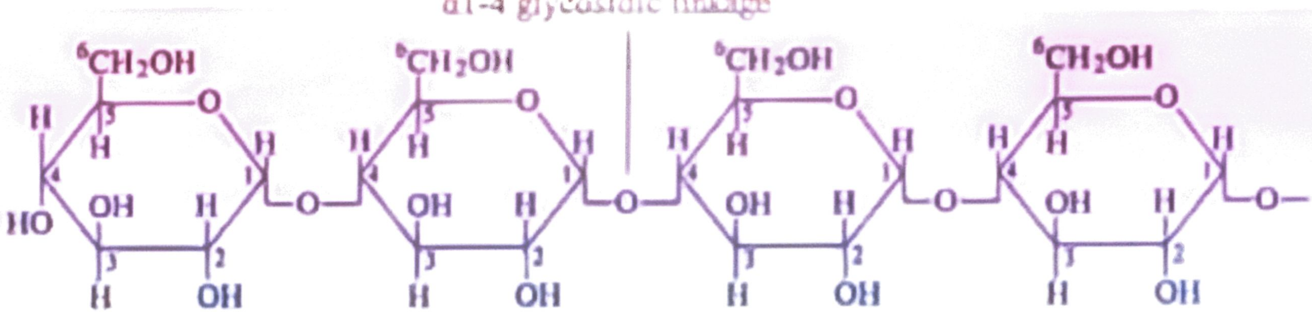
- Starch is used for energy storage in plants.
  - it can be separated into two fractions; amylose and amylopectin; each on complete hydrolysis gives only D-glucose.
  - **Amylose** is composed of continuous, unbranched chains of up to 4000 D-glucose units joined by α-1,4-glycosidic bonds.
  - **amylopectin** is a highly branched polymer of D-glucose; chains consist of 24-30 units of D-glucose joined by α-1,4-glycosidic bonds and branches created by α-1,6-glycosidic bonds.



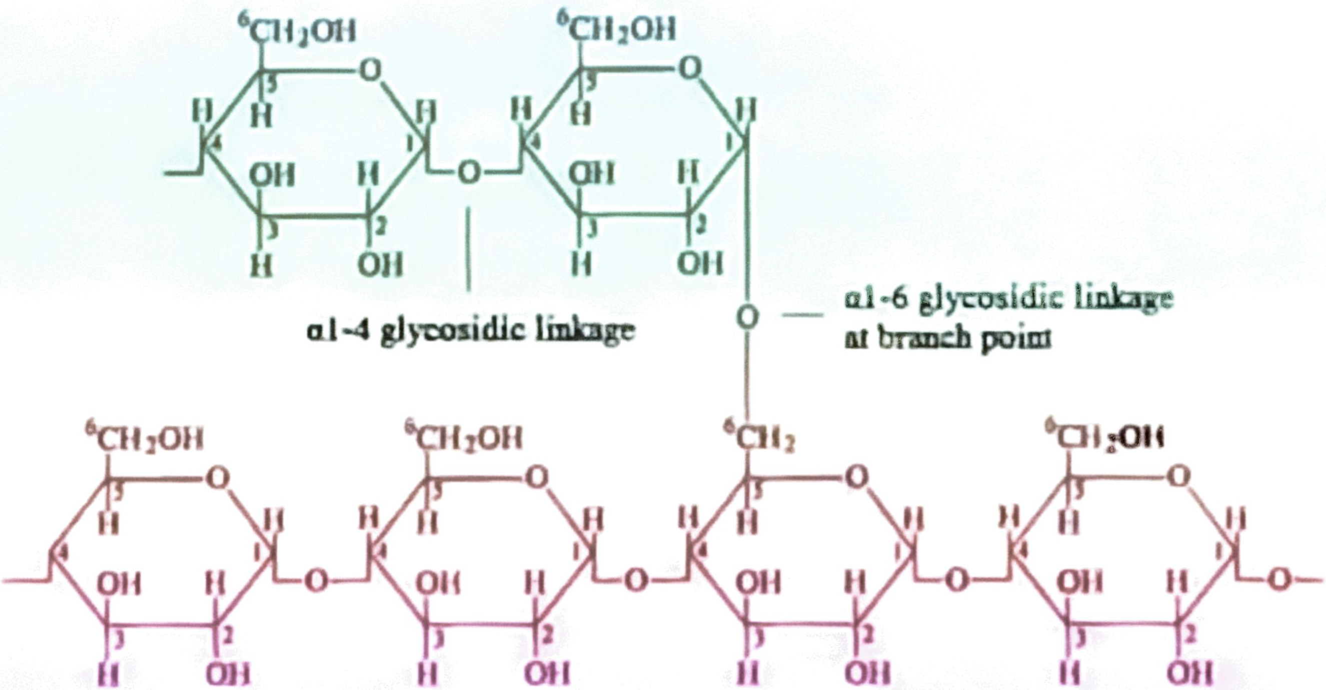
Amylose



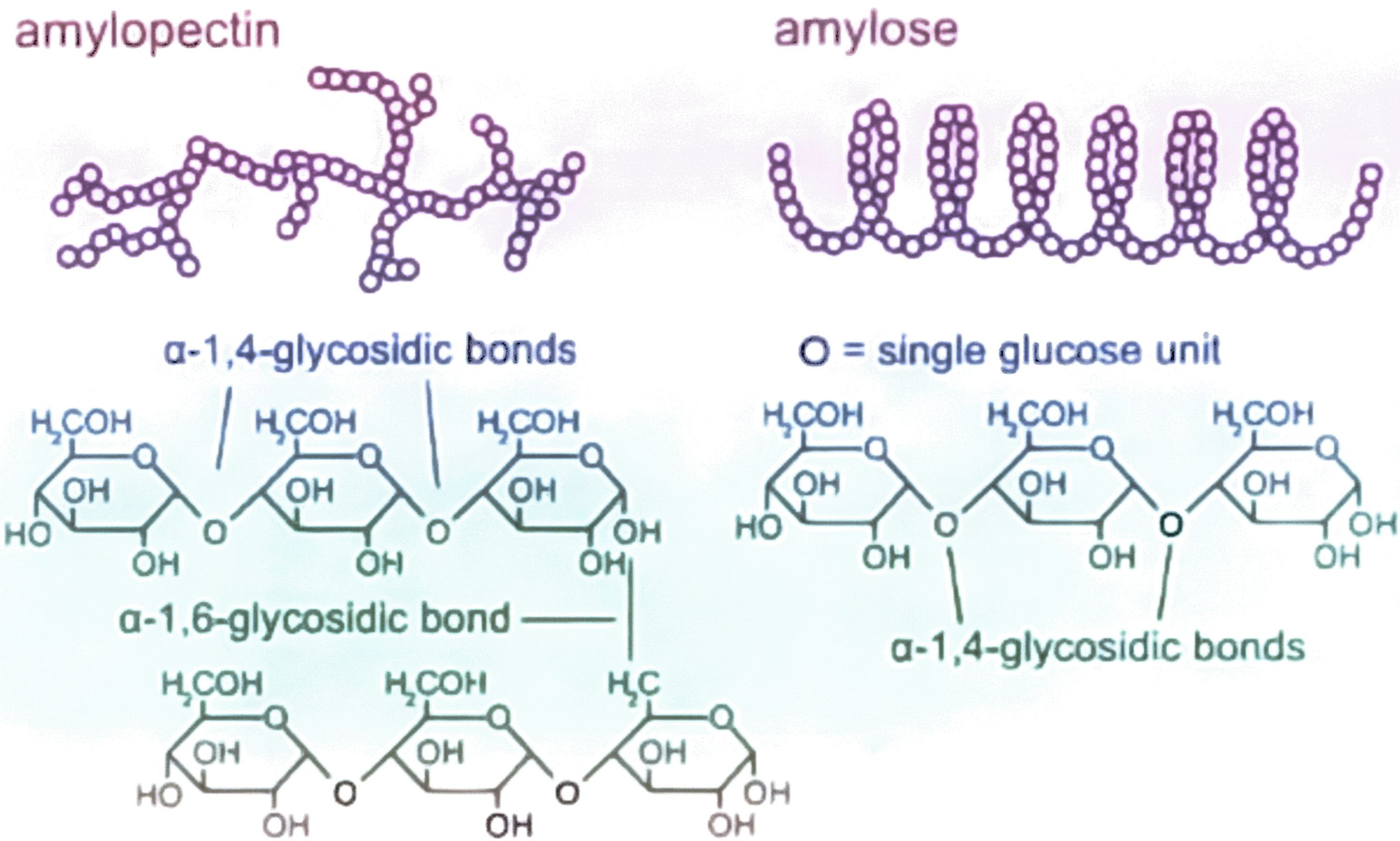
Amylopectin



Amylose



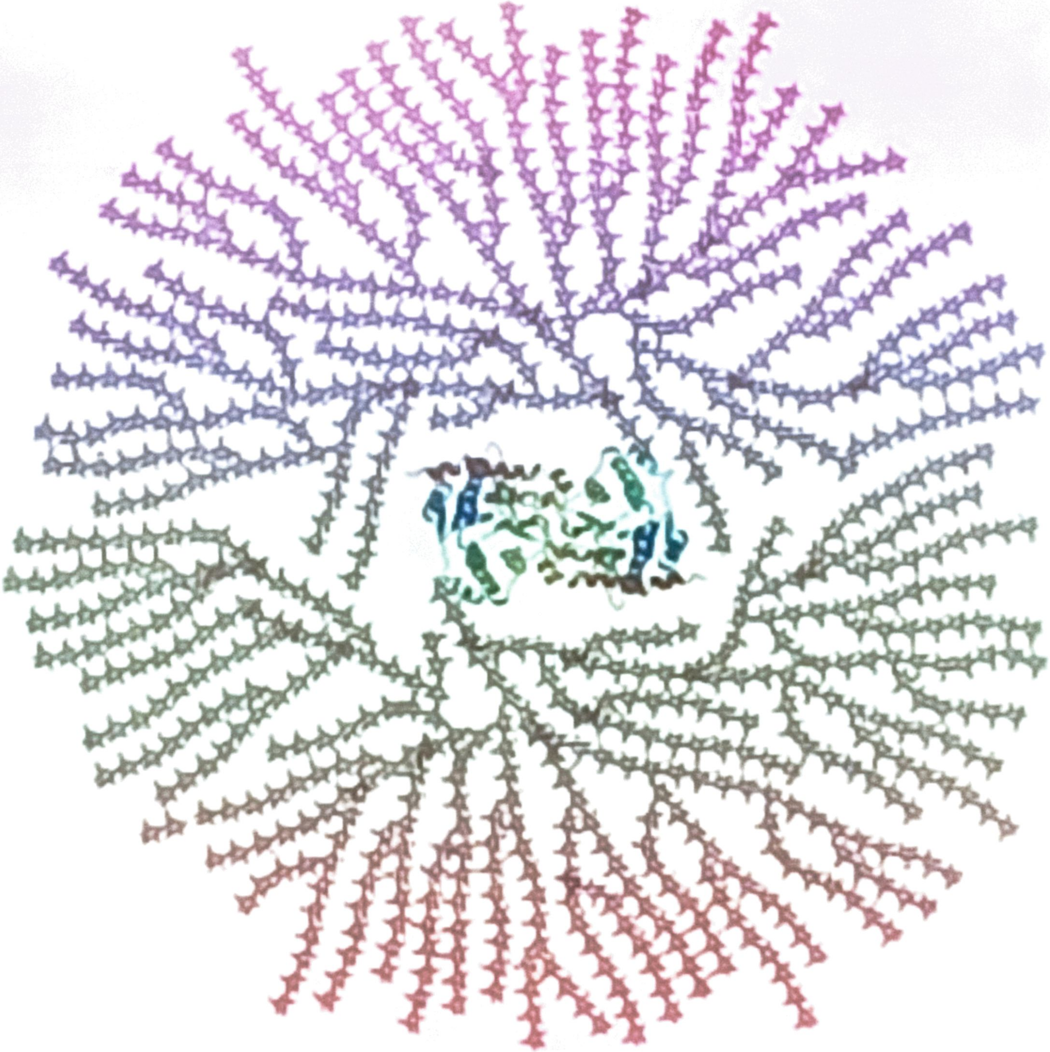
Amylopectin


amylopectin
amylose
α-1,4-glycosidic bonds
O = single glucose unit
α-1,6-glycosidic bond
α-1,4-glycosidic bonds

Homopolysaccharaides: B. Glycogen

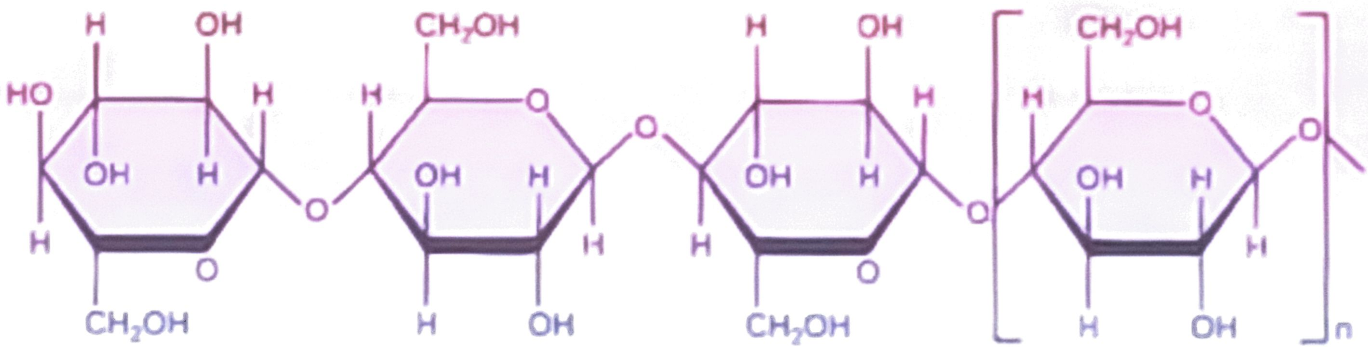
- Glycogen is the reserve carbohydrate for animals.
  - like amylopectin, glycogen is a nonlinear polymer of D-glucose units joined by α-1,4- and α-1,6-glycosidic bonds bonds.
  - the total amount of glycogen in the body of a well-nourished adult is about 350 g (about 3/4 of a pound) divided almost equally between liver and muscle.

A core protein of glycogenin is surrounded by branches of glucose units. The entire globular granule may contain around 30,000 glucose units

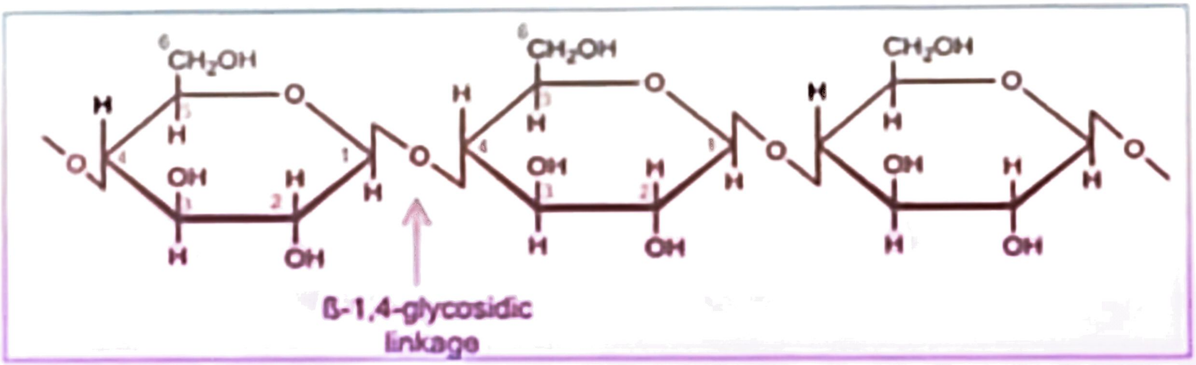


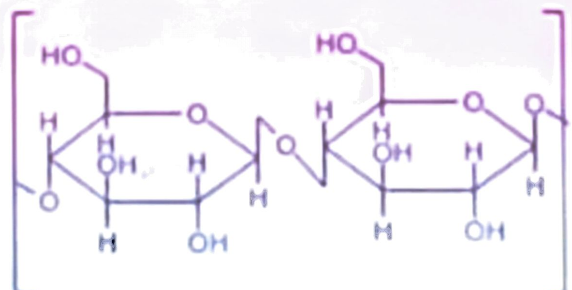
Homopolysaccharaides: C. Cellulose

- Cellulose is a linear polymer of D-glucose units joined by β-1,4-glycosidic bonds.
    - it has an average molecular weight of 400,000 g/mol, corresponding to approximately 2800 D-glucose units per molecule.



In order to make beta 1-4 glycosidic bonds, every alternate glucose molecule in cellulose is inverted. The hydroxyl group of the next carbon is directed upwards, and that of carbon 4 is directed downward.

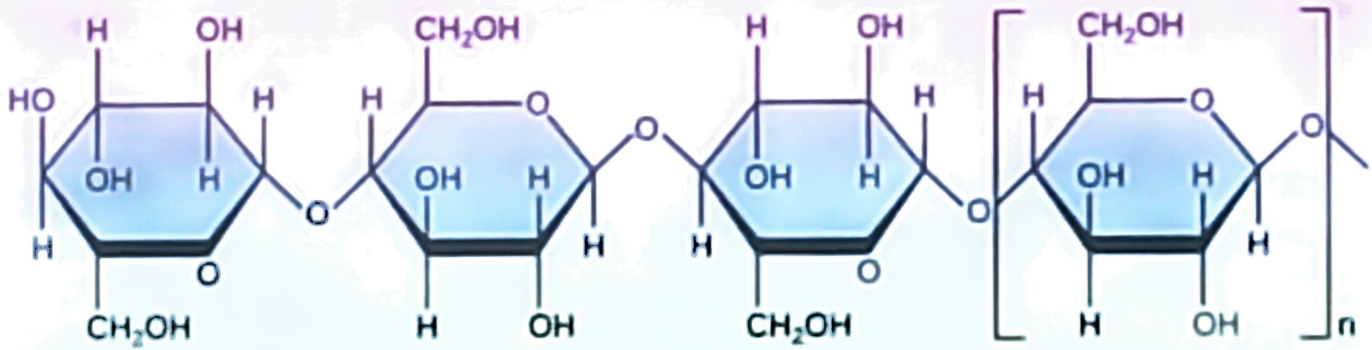




Cellulose

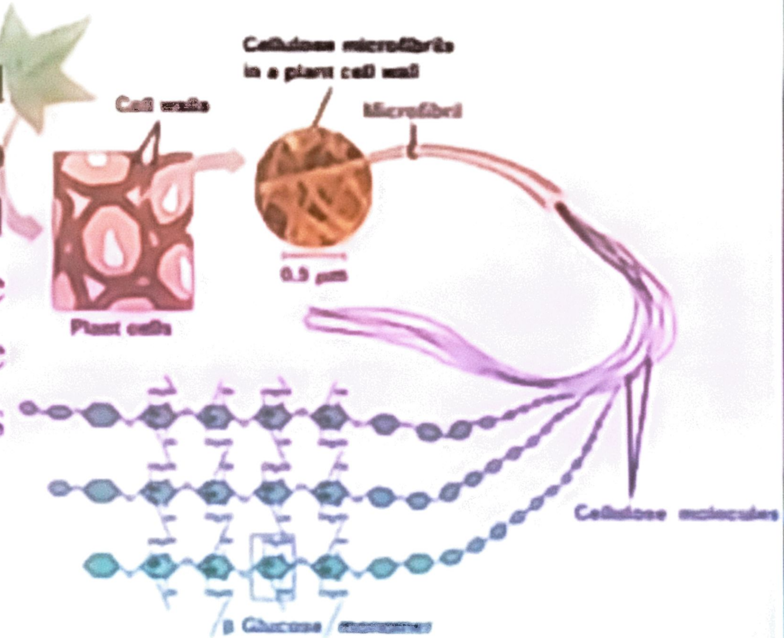
Now, to make a beta 1-4 glycosidic bond, one of these molecules should be inverted so that both the hydroxyl groups come in the same plane. This is the reason for the inversion of every alternate glucose molecule in cellulose.

Cellulose is an unbranched molecule.



Cellulose is an unbranched molecule. The polymeric chains of glucose are arranged in a linear pattern. Unlike starch or glycogen, these chains do not undergo any coiling, helix formation or branching. Rather, these chains are arranged parallel to each other.

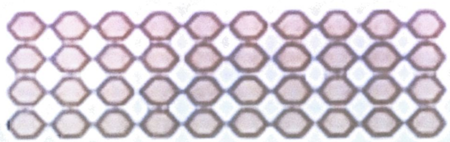
The hydrogen bonds are formed between these chains due to hydrogen atoms and hydroxyl groups which firmly hold the chains together. This results in the formation of cellulose microfibrils that are firm and strong.



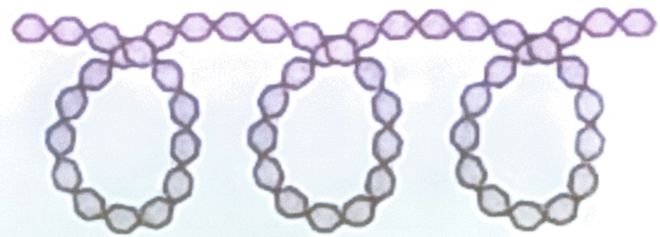
Cellulose is the most abundant carbohydrate present in nature

POLYSACCHARIDES

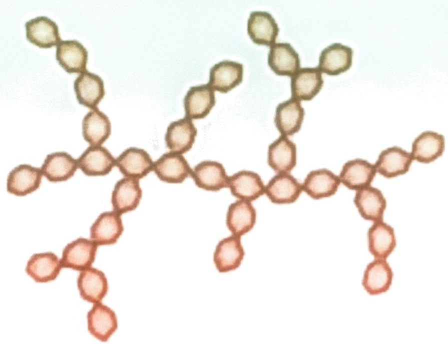
Cellulose



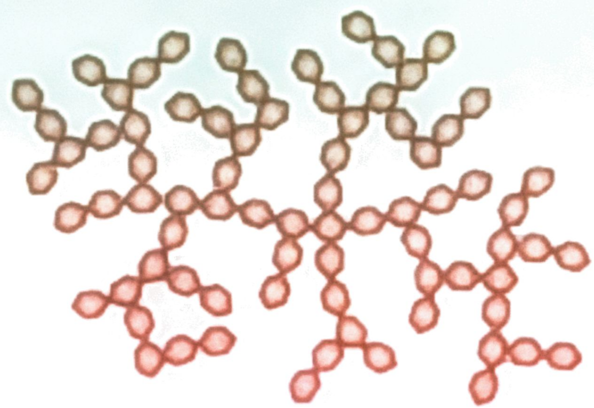
Amylose

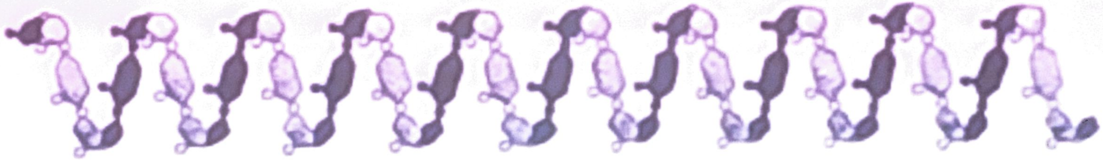


Amylopectin

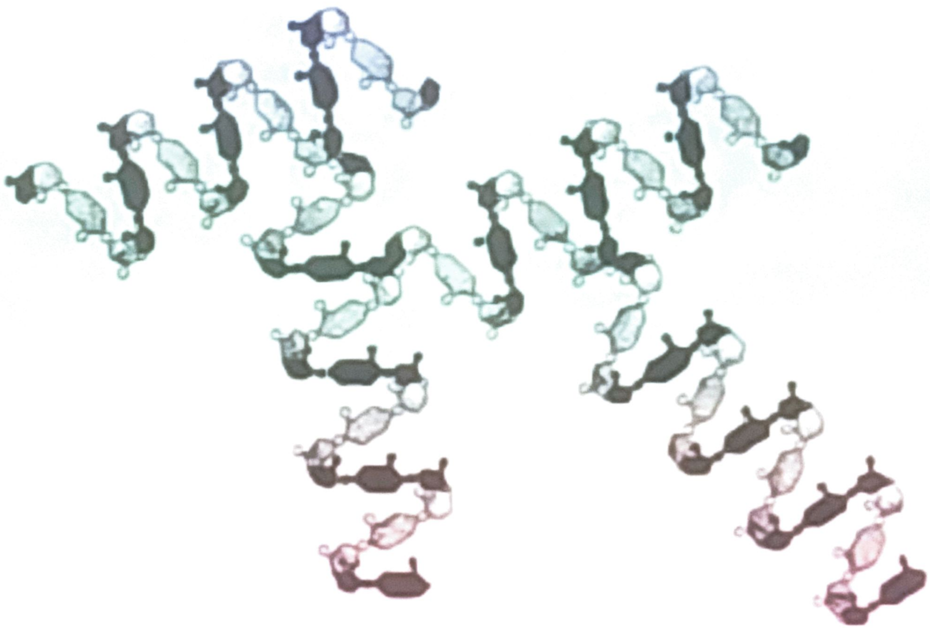


Glycogen



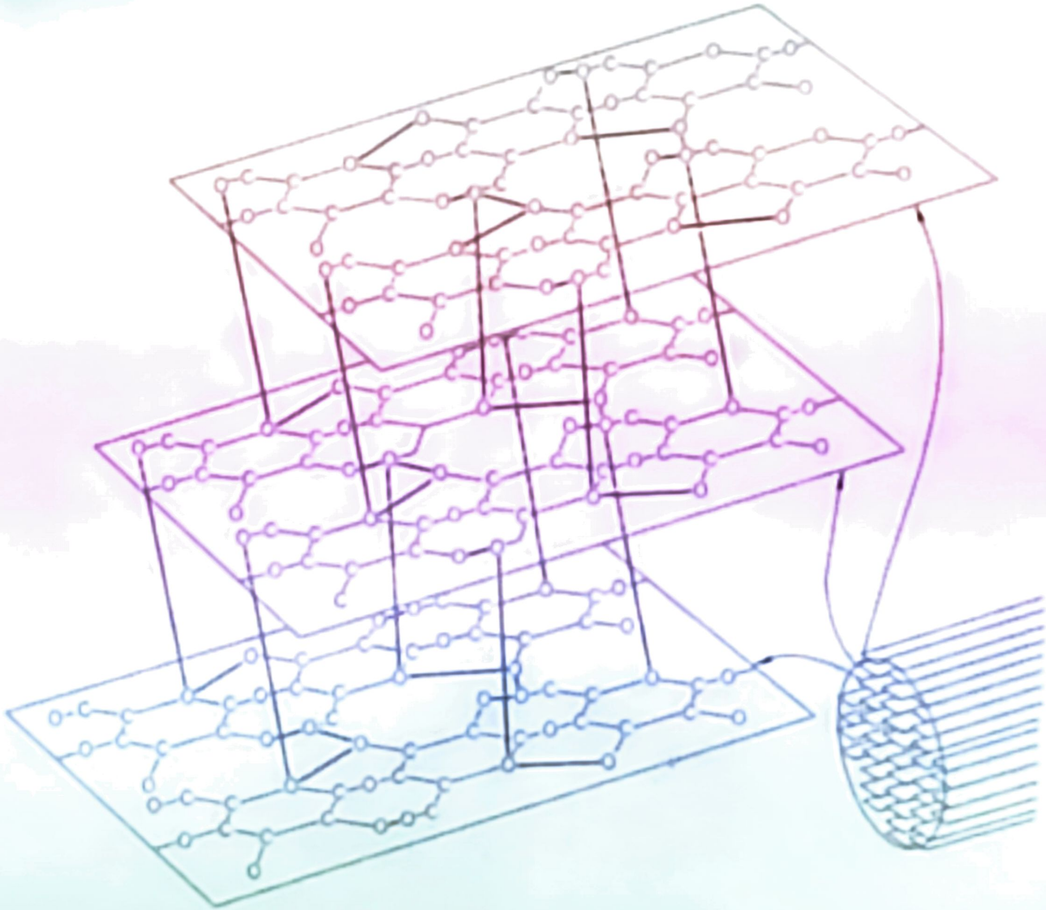


AMYLOSE



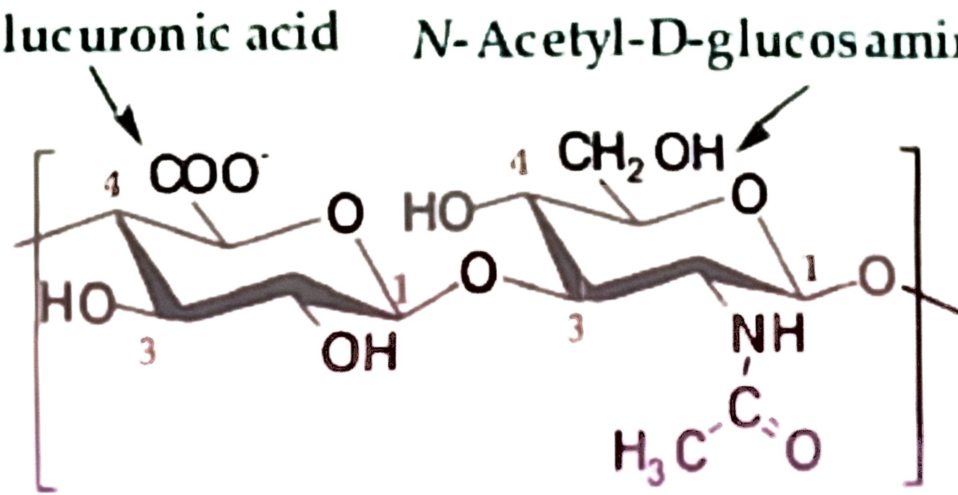
AMYLOPECTIN

Model of cellulose molecules in a microfibril



25.6 Heteropolysaccharides: A. Acidic

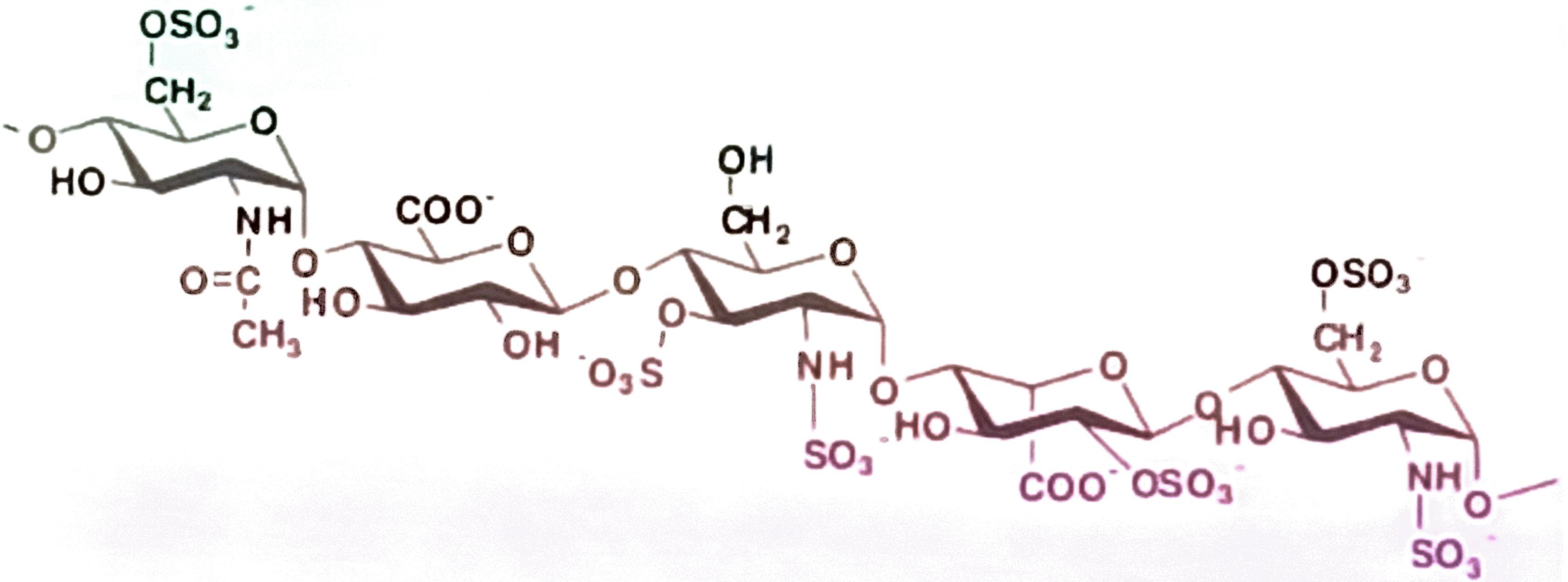
- **Hyaluronic acid:** an acidic polysaccharide present in connective tissue, such as synovial fluid and vitreous humor.



The repeating unit of hyaluronic acid

B. Acidic Heteropolysaccharides

- **Heparin:**
  - its best understood function is as an anticoagulant.
  - following is a pentasaccharide unit of heparin.



Dietary carbohydrates

- Starch
- Sucrose
- Glucose and fructose
- Lactose
- Cellulose
- Other plant polysaccharides

Digestible

Non-digestible by humans

Only monosaccharides are absorbed into the bloodstream from the gut.

Digestion of carbohydrates involves their hydrolysis into monosaccharides

Digestive Enzymes

Enzymes for carbohydrate digestion

Enzyme	Source	Substrate	Products
α-Amylase	Salivary gland Pancreas	Starch, glycogen	Oligosaccharides
Dextrinase	Small intestine	Oligosaccharides	Glucose
Isomaltase	Small intestine	α-1,6-glucosides	Glucose
Maltase	Small intestine	Maltose	Glucose
Lactase	Small intestine	Lactose	Galactose, glucose
Sucrase	Small intestine	Sucrose	Fructose, glucose

Lactase deficiency produces lactose intolerance

Blood glucose concentrations

Measured in mmol/L = mM or in mg/dL

Conversion factor: 1 mM = 18 mg/dL

Normal plasma glucose concentrations roughly
3.9 – 8.3 mM

Hypoglycemia: < 2.2 mM

Diabetes: > 7.0 mM (fasting)
> 11.1 mM 2 h after ingestion of 75 g glucose

LIPIDS

CLASSIFICATION OF LIPIDS

- Simple lipid
- Complex lipid
- Derived lipids

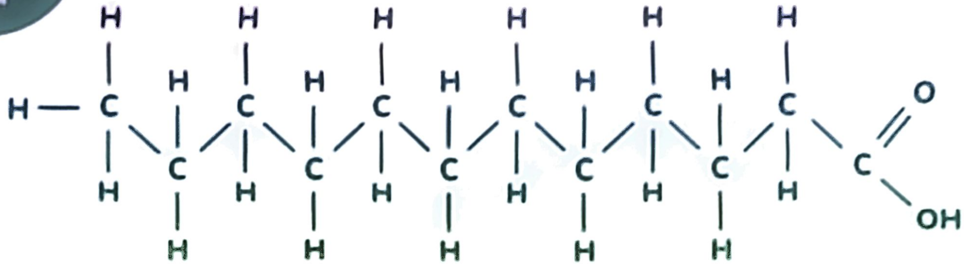
SIMPLE LIPIDS

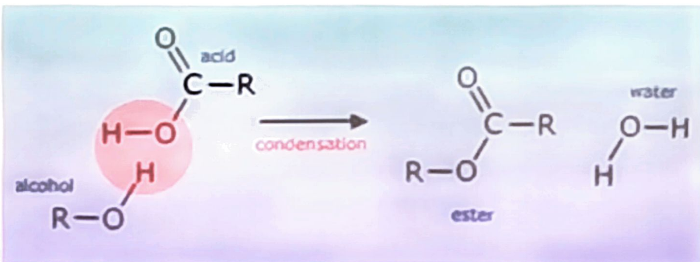
They are esters of FA with various alcohols

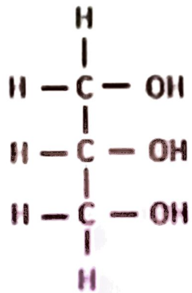
Depending up on the type of alcohols these are subclassified as

Neutral fats or oils	Waxes
Alcohol is GLYCEROL	Alcohol is other than glycerol

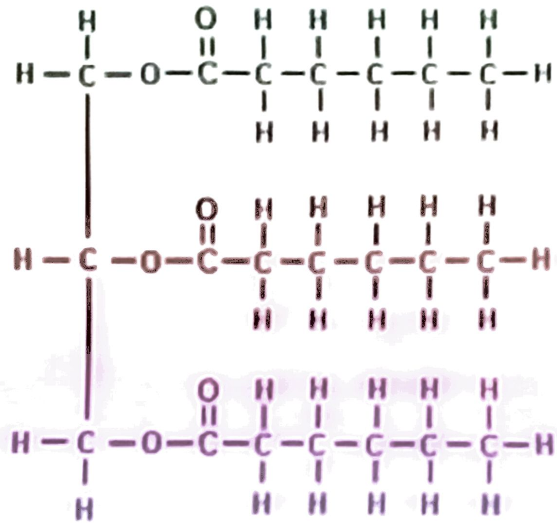
Fatty Acid





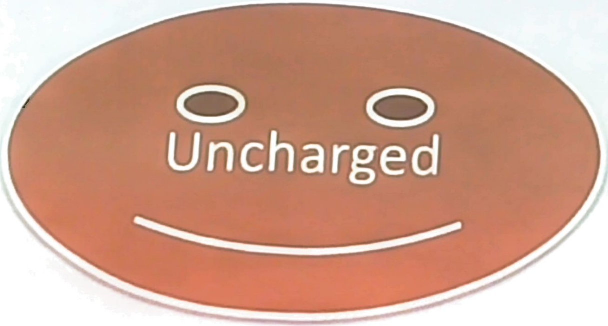


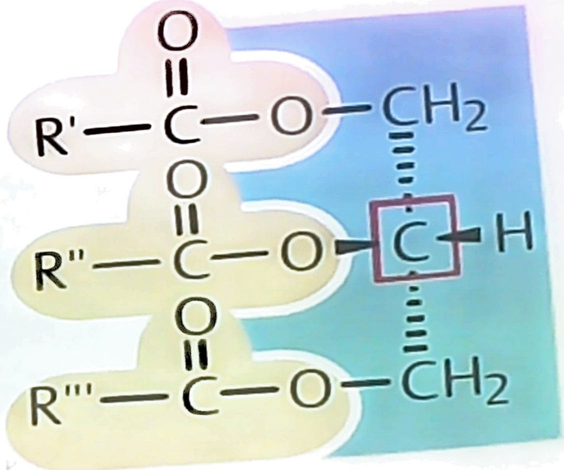
Glycerol



NEUTRAL FATS OR OILS

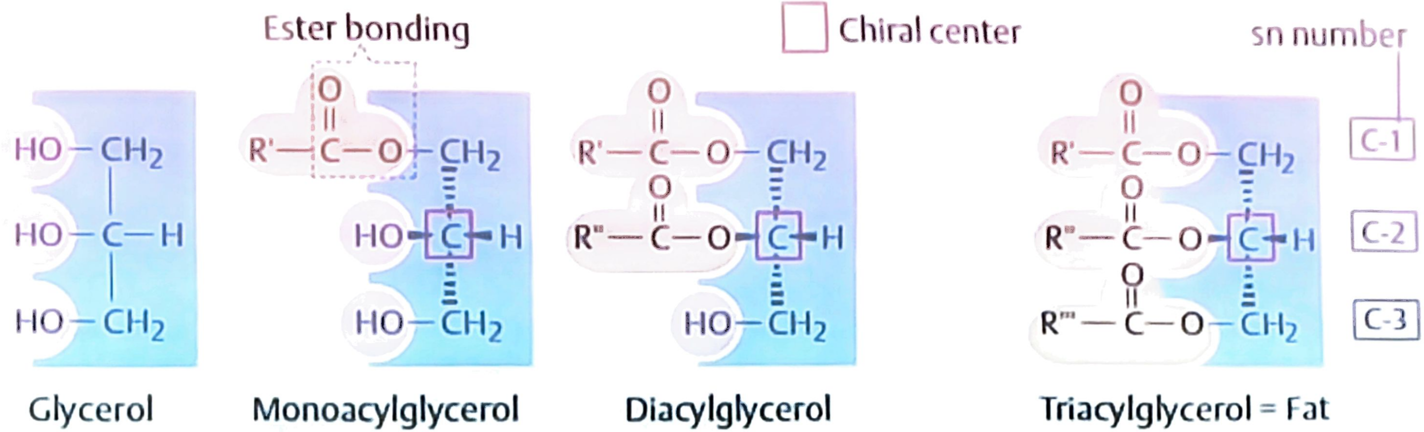
Esters of FA with alcohol GLYCEROL

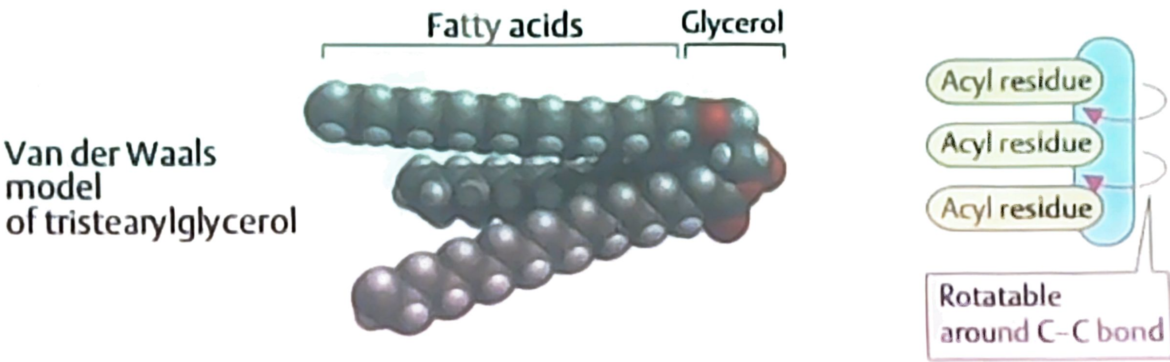




Triacylglycerol = Fat

B. Structure of fats





WAXES

Esters of FA with higher molecular weight monohydric alcohols

EXAMPLES

- ✓ Lanolin
- ✓ Beeswax
- ✓ Whale sperm oil

The true waxes are simple lipids which are the ester of fatty acids with acetyl alcohol or other high molecular weight alcohols. Fatty acids such as Myristic and Palmitic are combined with alcohols that contain from 12 to 30 carbon atoms.

Common naturally occurring waxes are beeswax of the honeycomb. Lanolin derived from wool is widely used as a base for many ointments and creams. Spermacetic derived from the sperm whale is used In cosmetics, pharmaceuticals and in the manufacture of candles.

$$\underbrace{CH_3(CH_2)_{14}-\overset{\overset{O}{\|}}{C}}_{\text{Palmitic acid}}-\underbrace{O-CH_2-(CH_2)_{28}-CH_3}_{\text{Triacontanol}}$$

Lanolin

The chemical formula of Paraffin waxes is $C_{20}H_{42}$. Its Melting point is 47 to 65-degree centigrade.

Wax

Paraffin

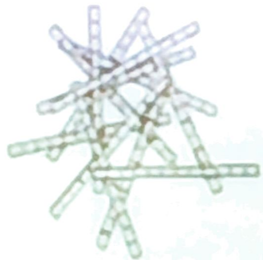
Iso-paraffin

Cycloparaffin

$$C_nH_{2n+2}$$

General Formula of Paraffin

Paraffin Wax



Microcrystalline Wax

COMPLEX LIPIDS

These are esters of FA with alcohol containing additional[prosthetic] groups.

Subclassified according to the type of prosthetic group

- Phospholipids
- Glycolipids
- Lipoproteins

COMPLEX LIPIDS

These are esters of FA with alcohol containing additional[prosthetic] groups.

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- Phospholipids
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